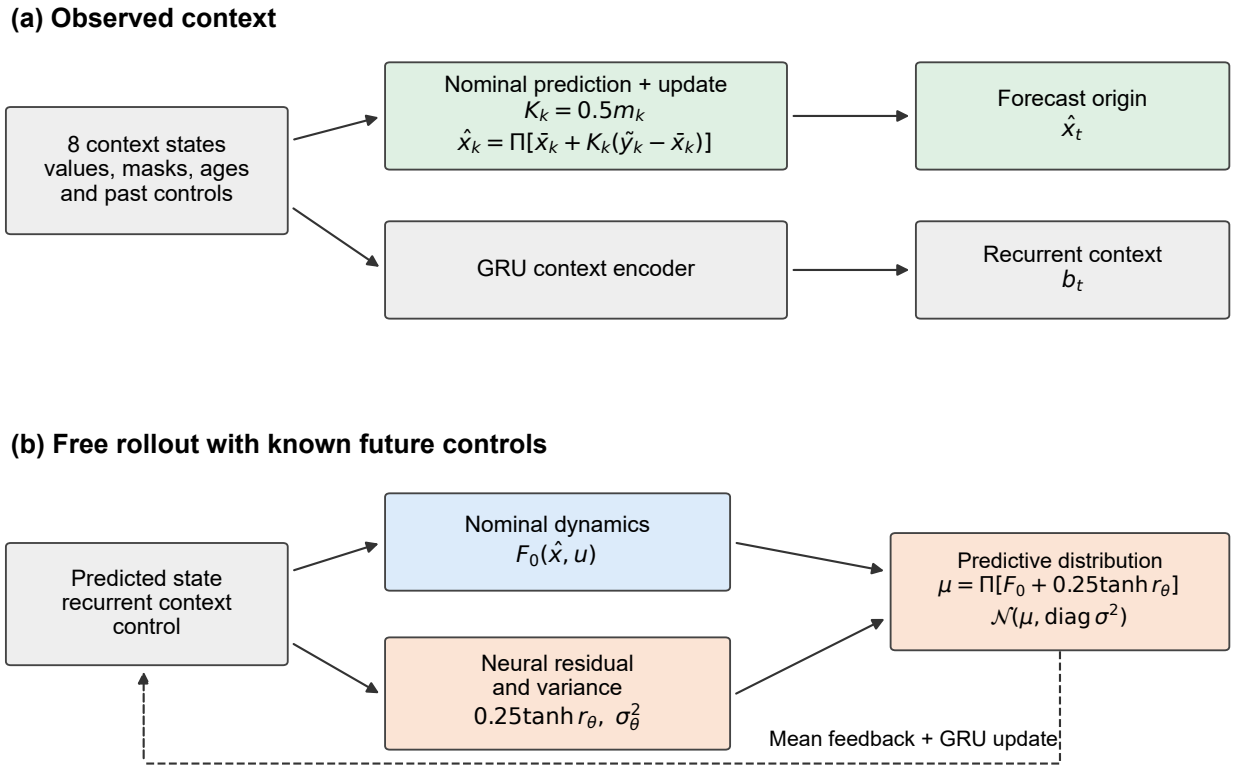

(a) Observed context
8 context states
values, masks, ages
and past controls
Nominal prediction + update
$K_k = 0.5m_k$
$\hat{x}_k = \Pi[\bar{x}_k + K_k(\tilde{y}_k - \bar{x}_k)]$
Forecast origin
$\hat{x}_t$
GRU context encoder
Recurrent context
b_t
(b) Free rollout with known future controls
Predicted state
recurrent context
control
Nominal dynamics
$F_0(\hat{x}, u)$
Neural residual
and variance
$0.25\tanh r_\theta,\ \sigma_\theta^2$
Predictive distribution
$\mu = \Pi[F_0 + 0.25\tanh r_\theta]$
$\mathcal{N}(\mu, \mathrm{diag}\,\sigma^2)$
Mean feedback + GRU update